



Chemical kinetic model reduction through species-targeted global sensitivity analysis (STGSA)

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ABSTRACT

The uncertainty of the rate coefficients of elementary reactions in detailed chemical kinetic mechanisms can affect the predicted species concentrations and global properties. The size of reduced mechanisms generated by reduction methods, such as directed relation graph (DRG), can also be greatly affected by the uncertainties in the rate coefficients, because these methods eliminate species by comparing contributions of the production rates of important species from other species. In order to obtain the optimum reduced mechanism in the uncertainty domain of reaction rate parameters, the species-targeted global sensitivity analysis (STGSA) method is proposed to measure the importance of species in kinetic mechanisms and to simplify them by considering the input uncertainties. The demonstrations are performed for the ethylene oxidation mechanism using the USC-Mech II with 111 species and the n-heptane oxidation mechanism using the JetSurf (version 1.0) with 194 species. It is found that STGSA can accurately identify the redundant global species and generate more compact reduced mechanisms compared with the DRG-based methods, resulting in a 33-species reduced mechanism for ethylene oxidation and a 37-species reduced mechanism for n-heptane oxidation. These reduced mechanisms show accurate predictions for the ignition delay times and laminar flame speeds over a wide range of operating conditions.

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1. Introduction

With numerical simulation assuming ever greater importance in the analysis of reactive flow characteristics and the corresponding implication in engine design [1], it is recognized that realistic simulation must necessarily require comprehensively accurate description of the chemical interactions from low to high-temperature chemistry in different combustion regimes for practical fuels, that involve large kinetic mechanisms with hundreds of species and thousands of reactions. For example, a detailed n-heptane mechanism [2] involves 1034 species and 4236 reactions, while detailed mechanisms for practical diesel and bio-diesel fuels could include nearly 3000 chemical species and 10,000 elementary reactions [3].

Clearly, 2D or 3D computational fluid dynamics simulation with such large detailed mechanisms cannot be achieved because of such excessively demanding computational load.

Recognizing the above constraints, it is imperative to develop skeletal mechanisms of less than 100 species that could retain their controlling chemical information and are also computationally efficient and robust, avoiding possible stiffness and numerical instabilities [4,5]. In the field of combustion [4,6], various reduction methods have been developed, such as sensitivity analysis (SA) [7,8], principal component analysis (PCA) [9–11], directed relation graph (DRG) [5,12,13], path flux analysis (PFA) [14,15], global pathway selection (GPS) [16], computational singular perturbation (CSP) [17–20], and tangential stretching rate (TSR) [21]. The reduction of the computation time can be substantial as it varies almost cubically with the number of species for the typical implicit solvers used in stiff systems.

A noted mechanism reduction algorithm for removing unimportant species is the DRG method which can abstract the couplings among the species and has been widely applied for many fuels.

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DRG has been subsequently improved by modifying the connection weight between pairs of species, leading to the further development of path flux analysis (PFA), directed relation graph with error propagation (DRGEP) and global pathway selection (GPS). These approaches aim to remove species by comparing contributions of the production rate of every species upon the production rate $\dot{\omega}_k$ of the k th major species. The important species determined by DRG-based reduction methods indicate that these species have non-negligible effects on certain species in the skeletal mechanism. However, not every species in the skeletal mechanism is of equal importance to the global parameters. As such, methods like sensitivity analysis are used to remove unimportant species from the skeletal mechanism, based on the worst-case induced error, and a series of simulations are necessary to identify all the unimportant species. Based on such sensitivity-based reduction approaches, additional reduction methods have been developed, e.g. DRG-aided sensitivity analysis (DRGASA) [22,23] and directed relation graph with error propagation and sensitivity analysis (DRGEP-ASA) [24]. Although these reduction methods combining DRG-based and sensitivity analysis can obtain very compact reduced mechanisms, they cannot be applied to mechanism reduction with consideration of the uncertainty. For many reactions in the detailed mechanisms, the reaction rate coefficients cannot be evaluated with high accuracy. This leads to the consideration of sensitivity as affected by the uncertainties in the reaction rate coefficients on the calculated species concentrations. Galassi et al. [20] indicated that overconfidence in the detailed mechanism can lead to misplaced focus on the tight error tolerances in the skeletal mechanism with respect to a faulty/uncertain baseline. Recognizing the above considerations, we present herein a species-targeted global sensitivity analysis (STGSA) aiming to measure the importance of species and simplify the combustion mechanism under the uncertainties in the reaction rate coefficients.

We next note that sensitivity analysis (SA) is an effective tool to gain insight into the complex reaction kinetic mechanisms and identify the effects of changes in the model input parameters on the output of the mathematical models. It facilitates modeling of detailed chemical mechanisms, identifies important reactions, and simplifies the reaction mechanisms, etc. [4,25,26]. For a homogeneous reaction system is written as:

$$\frac{dc}{dt} = F[c; [k]], \quad (1)$$

where c is the vector of species concentrations, $[k]$ the set of reaction rates, and F a complicated function of the concentrations and reaction rates. While the sensitivity of reaction systems to uncertainties in the reaction rates can be easily obtained by perturbing the reaction rates, the sensitivity of reaction systems to uncertainties in the species concentrations are difficult to measure. Consequently, global sensitivity analysis of reaction systems to species still remains a challenge. One way to measure the importance of a species is to evaluate the simultaneous effects of all the reactions associated with such species. Although local sensitivity analysis (LSA) known as one-variable-at-a-time (OFAT) can be used to quickly calculate sensitivity coefficients, it cannot evaluate the joint effects of multi-variables on the outputs. Many global sensitivity analysis (GSA) tools can be employed to analyze or reduce complex reaction systems, e.g. Morris [27,28], HDMR [29–31], Sobol [32] etc. At present, these techniques are mainly employed to measure the importance of reactions and reduce the unimportant reactions. In this work, the present STGSA approach based on Sobol index is designed to quantitatively measure the importance of species and remove the global redundant species. In order to show the differences between present and previous works, a summarization of Sobol-based methods for kinetic analysis and reduction of chemical kinetic systems is shown in Table 1. These researches can be

Table 1

The applications of Sobol-based methods on analysis and reduction of chemical kinetic systems.

Type	Feature	Authors
A	Validate the reliability of sensitivity indices obtained by other GSA tools.	T. Ziehn et al. [33], An et al. [34], Saltelli et al. [35]
B	Reduce computational cost.	S. Li et al. [36], V. Todorov et al. [37]
C	Generate a reduced reaction mechanism by measuring the importance of reactions.	L.F. Nurislamova et al. [38], L.F. Nurislamova et al. [39], M.S. Okino et al. [40]

divided into 3 types. For type A, Sobol is employed to verify whether the sensitivity indices obtained by other GSA tools are correct or not. The type B combines Sobol and other technologies in order to reduce the computational cost. And type C is for mechanism reduction using the Sobol indices.

2. Methodologies

2.1. Sobol sensitivity technique

Sobol sensitivity technique is based on the analysis of variance (ANOVA) decomposition. It decomposes the variance of the output of the model into fractions which can be attributed to inputs. Any square-integrable model can be considered as a function $Y = f(x)$, where x is a vector of n uncertain model inputs ($x_1, x_2, \dots, x_n$) defined in the unit hypercube, i.e. $x_i \in [0, 1]$ for $i = 1, 2, \dots, n$. Then $f(x)$ can be decomposed in the form [41]:

$$Y = f_0 + \sum_{i=1}^n f_i(x_i) + \sum_{1 \leq i < j \leq n} f_{ij}(x_i, x_j) + \dots + f_{1,2,\dots,n}(x_1, x_2, \dots, x_n) \quad (2)$$

There are many ways of decomposing f , but for independent inputs there exists a unique decomposition in orthogonal terms. And analysis of the variance representation of $f(x)$ is based on satisfying the following condition:

$$\int_0^1 f_{i_1, \dots, i_s}(x_{i_1}, \dots, x_{i_s}) dx_k = 0 \quad \text{for } k = i_1, \dots, i_s, \\ 1 \leq i_1 < \dots < i_s \leq n, \quad (3)$$

This leads to definitions of the terms of the functional decomposition in terms of conditional expected values:

$$\begin{cases} f_0 = \int f(x) dx \\ f_i(x_i) = \int f(x) \prod_{k \neq i} dx_k - f_0 \\ f_{ij}(x_i, x_j) = \int f(x) \prod_{k \neq i, j} dx_k - f_0 - f_i(x_i) - f_j(x_j) \\ \vdots \end{cases} \quad (4)$$

Then further assume that $f(x)$ is square-integrable, the functional decomposition can be squared and integrated to yields:

$$\int f^2(x) dx - f_0 = \sum_{s=1}^n \sum_{i_1 < \dots < i_s} \int f_{i_1, \dots, i_s}^2 dx_{i_1} \dots dx_{i_s} \quad (5)$$

where the left hand side is equal to the variance of Y , and the terms of the right hand side are variance terms. This finally leads to the decomposition of variance expression:

$$\text{Var}(Y) = \sum_{i=1}^n V_i + \sum_{i < j} V_{ij} + \sum_{i < j < l} V_{ijl} + \dots + V_{1,2,\dots,n}, \quad (6)$$

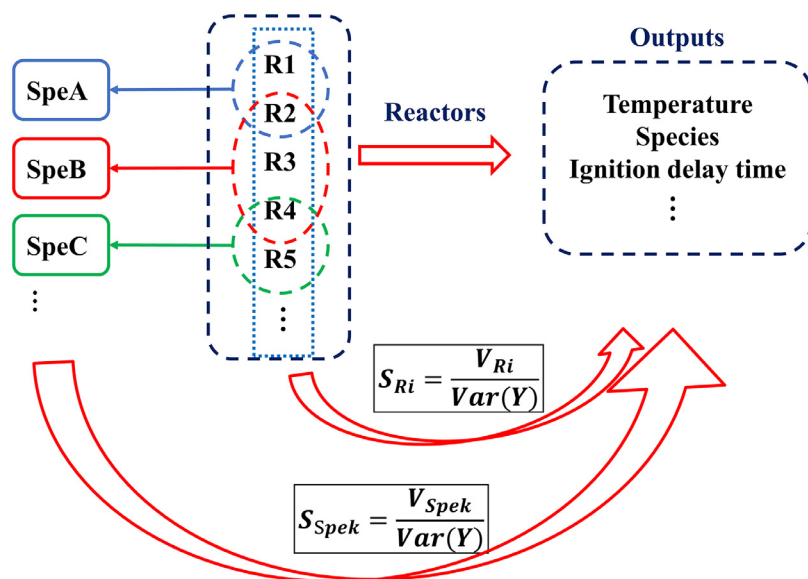


Fig. 1. Schematic of sensitivity analysis on species. R_i indicates the i th reaction, Spe_k indicates the k th species and Reactors indicates the combustion reactor models such as a perfectly stirred reactor (PSR). Such a figure shows a chemical species A involves in reactions R1 and R2, and species B involves in reactions R2, R3 and R4. $S_{Ri} = \frac{V_{Ri}}{Var(Y)}$ is employed to evaluate the importance of a reaction. $S_{Spek} = \frac{V_{Spek}}{Var(Y)}$ is proposed in this work to measure the importance of species.

where V_i is the variance of $f_i(x_i)$, V_{ij} the variance of $f_{ij}(x_i, x_j)$ and so on. The Sobol sensitivity indices of inputs are then defined as:

$$S_{i_1, \dots, i_s} = \frac{V_{i_1, \dots, i_s}}{Var(Y)}, \quad (7)$$

Sobol provides a Monte Carlo strategy to compute indices of any order, and it can evaluate the total sensitivity indices in a more compact way and it gives as:

$$S_{x_i} = 1 - S_{\sim x_i} \quad (8)$$

where $S_{\sim x_i}$ is a sensitivity index of all inputs but input x_i and can be easily calculated by Sobol without computing the $2^n - 1$ terms of the n -input model [42]. In this work the importance of a species is evaluated by total sensitivity index.

2.2. Sobol sensitivity analysis on chemical species

Sensitivity analysis is a crucial tool for insights into chemical kinetics systems. Many research have been performed in the global sensitivity analysis (GSA) of complex reaction systems aiming to measure the importance of reactions based on Eq. (7) [33,36,43,44]. In this work, a method of evaluating the importance of species based on Sobol is presented.

A gas-phase kinetics input file not only gives the rate coefficients for each reaction but it also demonstrates the connection between a species and the related reactions. A species can be involved in many reactions, and the rates of these reactions can directly affect such species. Thus, the sensitivity index of a chemical species can be estimated by measuring the joint effects of the reactions associated with such species. Figure 1 demonstrates how to measure the sensitivity index of a chemical species.

From Eq. (7) the first-order sensitivity indices for a reaction can be calculated with the formula:

$$S_{Ri} = \frac{V_{Ri}}{Var(Y)} \quad (9)$$

where V_{Ri} is the variance induced by the i th reaction, $Var(Y)$ the variance induced by all reactions, and S_{Ri} the sensitivity indices of the i th reaction. Such an equation is widely used to evaluate the

importance of reactions and yields insights into a complex combustion reaction system as an uncertainty factor (UF) for each reaction.

The importance of species cannot be directly evaluated because a chemical species concentration is not explicitly expressed and the uncertainty factor for each species is not known. On the other hand, the importance of species can be indirectly measured by treating its related reactions as whole. For example, in Fig. 1 the sensitivity indices of species A can be estimated by measuring the joint effects of reactions 1 and 2 because species A is only involved in reactions 1 and 2. In the same way, reactions 2, 3, and 4 are considered as a cluster when the importance of species B is evaluated. The formula for evaluating the importance of species is:

$$S_{Spek} = \frac{V_{Spek}}{Var(Y)} \quad (10)$$

where V_{Spek} is the variance induced by species k , $Var(Y)$ the variance induced by all species, and S_{SpeA} the sensitivity indices of species A. Because species A is involved in reactions R1 and R2, based on Eq. (7), Eq. (10) can be expressed as:

$$S_{R1,R2} = \frac{V_{R1,R2}}{Var(Y)} \quad (11)$$

Eq. (11) gives the formula of evaluating the importance of species under the uncertainties in the reaction rates. Sobol can easily obtain the sensitivity index of S_{SpeA} without calculating the sensitivity indexes for S_{R1} and S_{R2} .

Figure 2 briefly describes the calculation procedure of the sensitivity indices using the Sobol technique and Refs. [42,45] provide detailed information for the calculation. Because the reaction remains uncertain, a range of possible values is assigned to each reaction rate. It is seen in Fig. 2 $N \times 2d$ sample matrix is firstly generated using the sampling technique of Sobol sequences, where N and d are the sample size and the numbers of inputs, respectively. Here, N and d are set as 4 and 5, respectively. The first d columns of the matrix are set as M , and the remaining of d columns are set as M' . Each row of M or M' , and each column of M , indicates a set of possible values for inputs. When the sensitivity index of input R1 is measured, the column of R1 of the matrix M is replaced by $R1'$ and the new matrix $M^{(R1)}$ is obtained. Then, the output values

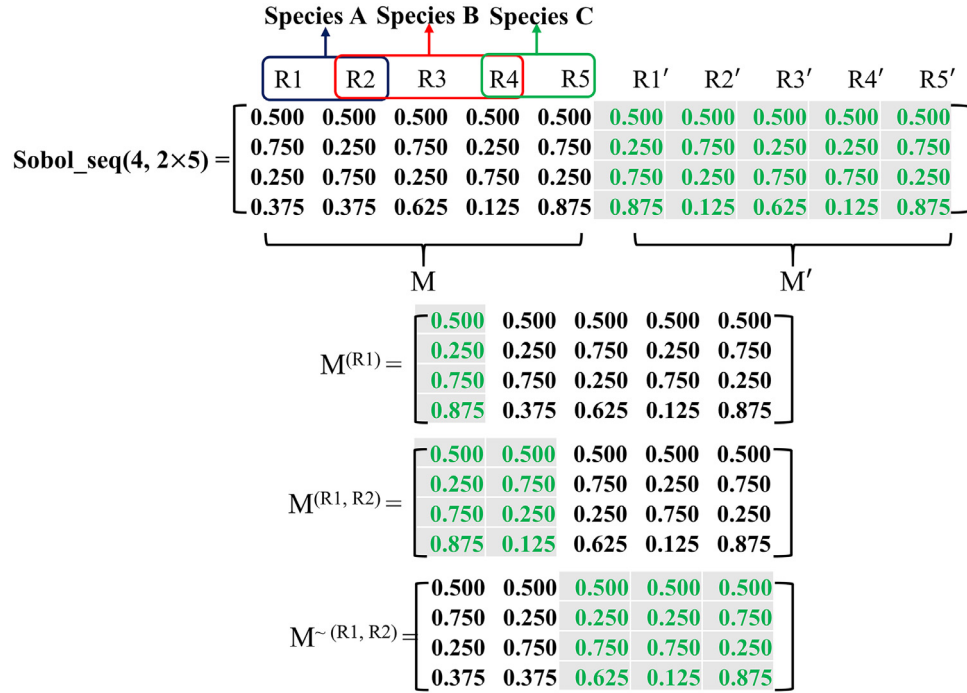


Fig. 2. An example of the calculation of sensitivity indices of species based on the Sobol method. Sobol_seq indicates the generation of the sample matrix using Sobol sequences. Ri indicates the *i*th input parameter, and Ri' is also the *i*th input parameter but its value is given by a matrix M'.

$f(M)$, $f(M')$ and $f(M^{(R1)})$ can be calculated by using the sample matrix M, M' and $M^{(R1)}$ respectively. Finally, the sensitivity index of an input R1 can be estimated as:

$$S_{R1} = \frac{V_{R1}}{\text{Var}(Y)}, \quad (12)$$

where $V_{R1} = \frac{1}{N} \sum_{j=1}^N f(M') (f(M^{(R1)}) - f(M))$. For reaction mechanism, the sensitivity index of species A which relates R1 and R2 can be calculated by the expression:

$$S_{\text{SpeA}} = S_{R1, R2} = \frac{V_{R1, R2}}{\text{Var}(Y)}, \quad (13)$$

where $V_{R1, R2} = \frac{1}{N} \sum_{j=1}^N f(M') (f(M^{(R1, R2)}) - f(M))$. The importance of species A is obtained by estimating the joint effects of R1 and R2. The total sensitivity index of species A is estimated by the expression:

$$S_{T_{\text{SpeA}}} = 1 - S_{\sim \text{SpeA}} = 1 - S_{\sim(R1, R2)} = 1 - \frac{V_{\sim(R1, R2)}}{\text{Var}(Y)}, \quad (14)$$

where $S_{T_{\text{SpeA}}}$ is a total sensitivity index of species A, $S_{\sim \text{SpeA}}$ is a sensitivity index of all inputs but species A, and $V_{\sim(R1, R2)} = \frac{1}{N} \sum_{j=1}^N f(M') (f(M^{\sim(R1, R2)}) - f(M))$. Thus, we use Eq. (14) to evaluate the importance of the chemical species A.

When the GSA tools are applied to evaluate the importance of species, the uncertainty factor (UF) for each reaction should be estimated and it may be found in a kinetic compilation, e.g. references [46,47]. Then a set of possible values for inputs are generated using the sampling technique of Sobol sequences, and a set of possible values for outputs can be subsequently acquired. Finally, the total sensitivity index of a chemical species is calculated from Eq. (14). Note that we discuss an extreme case that a chemical reaction mechanism involves the part of reactions with negligible uncertainties. In such case, when a species is

associated with all reactions with negligible uncertainties, such species cannot be evaluated using the present method. And reduction of such mechanism under uncertainty should involve both global and local sensitivity-based reduction methods because local sensitivity-based method is suitable for reactions with negligible uncertainties and global sensitivity-based method is appropriate for reactions with a certain degree of uncertainty. In this work, we consider the normal chemical reaction mechanism under uncertainties in the reaction rates.

Since STGSA can determine how a species contributes to an output behavior, it is suitable for the identification of unimportant species and mechanism reduction. Moreover, global sensitivity analysis of species can be a more efficient evaluation of a chemical kinetic mechanism compared to the global sensitivity analysis of reactions, because the number of reactions is usually five times more than the number of species [48]. In this work, JetSurf (version 1.0) mechanism which involves 194 species and 1459 reactions will be simplified by STGSA for n-heptane oxidation. In this case, global sensitivity analysis of species for such mechanism is much easy to implement compared to global sensitivity analysis of reactions. Hence, the STGSA approach can quickly obtain the results and make the GSA tool more widely applicable in reaction mechanism analysis.

2.3. Computational details

The demonstration of evaluating the present STGSA capability is performed for two combustion mechanisms including the high-temperature oxidation of ethylene and n-heptane using USC II [49] and JetSurf (Version 1.0) [50] mechanisms, respectively. The uncertainty factors of reaction rates for these two mechanisms have been measured in Refs. [50,51] and the combustion characteristics of these two fuels have attracted much attention in reacting flow analyses. Uncertainty factor (UF) is used for evaluation of a reaction rate uncertainty, defined as:

$$UF = \frac{k_0}{k_{min}} = \frac{k_{max}}{k_0} \quad (15)$$

where k_0 is the nominal value of a reaction rate, and k_{min} , k_{max} are the upper and lower bounds of the reaction rate, respectively. Then the samples which are uniformly distributed under uncertainties in the reaction rates are generated by Sobol sequences. Note that the selection of the probability density distributions (PDFs) of samples can impact on the resulting sensitivity indices for species. However, the different types of PDFs have hardly effects on the important reaction rankings. Thus, the unimportant species identified by the present method are hardly affected by the selection of PDFs.

Before evaluating the importance of species for different test cases, the required minimum sample sizes are first evaluated in an operating condition. Simulations with different sample sizes for each mechanism are conducted under temperature of 1000 K, pressure of 1.0 and equivalence ratio of 1.0 in order to determine the required minimum sample sizes for obtaining the reliable results of sensitivity indices.

In evaluating the importance of species, 27 target operating conditions are considered as the data source, with pressure of $P = [1 \text{ atm}, 5 \text{ atm}, 30 \text{ atm}]$, equivalence ratio of $\varphi = [0.5, 1.0, 1.5]$, and initial temperature of $T = [1000 \text{ K}, 1300 \text{ K}, 1600 \text{ K}]$. Based on Eq. (14), the sensitivity indices for each species at different operating conditions can be obtained, and the maximum sensitivity index for species under all operating conditions is used to evaluate their importance and hence their ranking.

In reaction mechanism reduction, the redundant species are gradually removed according to its sensitivity indices. The maximum relative error in ignition delay time between predictions of the reduced and detailed mechanisms is employed to evaluate the accuracy of the reduced mechanism, defined as:

$$\delta = \max_{k \in [Z]} \frac{|\tau_{det}^k - \tau_{skel}^k|}{\tau_{det}^k}, \quad (16)$$

where τ_{det}^k and τ_{skel}^k are the ignition delay times predicted using the detailed and reduced mechanisms respectively, k is the initial condition, and Z the set of all initial conditions. In this work the maximum relative error for the reduced mechanism is set at 20%.

3. Results and discussion

In this section, we shall first discuss the effects of the uncertainties in the rate coefficients on the chemical mechanism reduction, and indicate that it is difficult for DRG-based reduction methods to identify the coupling species from chemical mechanism under consideration of the uncertainties in the reaction rate coefficients. STGSA is then employed to simplify the USC II and Jet-Surf (Version 1.0) mechanisms for ethylene and n-heptane, respectively. The global redundant species identified by STGSA may also be unimportant species for the nominal values within the reaction mechanism. Thus, we also compare the present STGSA method and the DRG-based methods in generating the reduced mechanism.

3.1. Mechanism reduction under uncertainty

The uncertainties in the rate coefficients can affect the calculated concentrations of all the species, and as such can significantly impact the degree of dependence between two species. With consideration of the uncertainties in the rates, the coupling species identified by DRG-based methods may remain very much uncertain. For instance, the sample size of 100 for ethylene mechanism is generated under uncertainties in the reaction rates, and these mechanisms are then simplified by DRGEP under a wide range of operating conditions shown in Section 2.3, resulting in the series of the reduced mechanisms with different number of species, and it is found that the number of coupling species vary widely among the resulting reduced mechanisms, shown in Fig. 3.

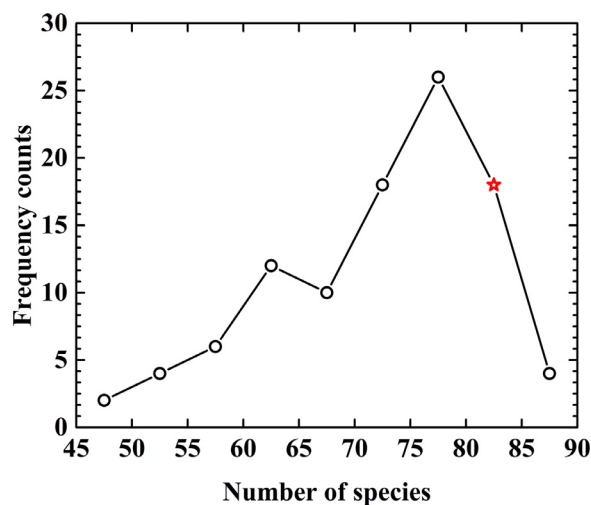


Fig. 3. Frequency counts of the number of species for the series of the reduced mechanisms. The reduced mechanisms for ethylene auto-ignition are obtained using DRGEP to reduce the chemical mechanisms generated by sampling of 100 sample under uncertainties in the reaction rates. Star means that the number of species in the reduced mechanism simplified using the original mechanism is within such interval.

Because of the uncertainties in the reaction rates, it is difficult to determine the coupling species using the DRG-based methods. It is also seen in Fig. 3 that there is significant difference in the number of species among these reduced mechanisms, and the maximum and minimum number of species in these reduced mechanisms are 86 and 49 respectively. Hence, it is necessary to develop the global reduction techniques to identify the global redundant species under uncertainties in the reaction rates.

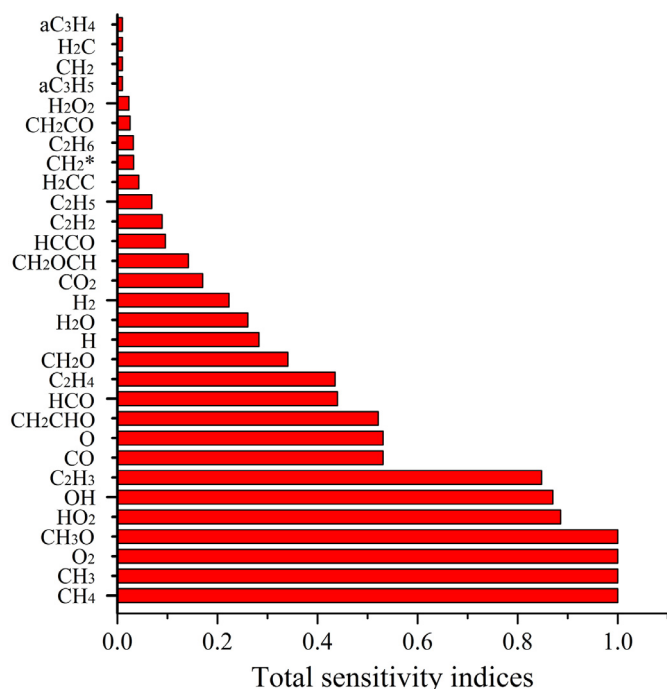
3.2. Mechanisms reduction for ethylene

The auto-ignition of ethylene/air mixture is calculated using the USC II mechanism with 111 species and 784 species. STGSA is applied to measure the importance of species for this chemical mechanism and generate the reduced mechanism of minimum size for ethylene auto-ignition. The minimum required sample size for achieving the correct ranking of the importance of species is measured, shown in Table 2. It is seen that the total sensitivity indices for five species are observed for different sample sizes, and small and large sensitivities for species are all accurately calculated with high sample size, and the sample size N may require 2048 in order to obtain a good estimate for the sensitivity indices. It can be also observed that very small sensitivity coefficients for some species are obtained. This means that these species can be considered as unimportant for ethylene auto-ignition because ignition delay times are not affected by the reactions associated with these species. In this case, the total sensitivity indices for all species are measured with the sample size of 2048 under a wide range of operating conditions.

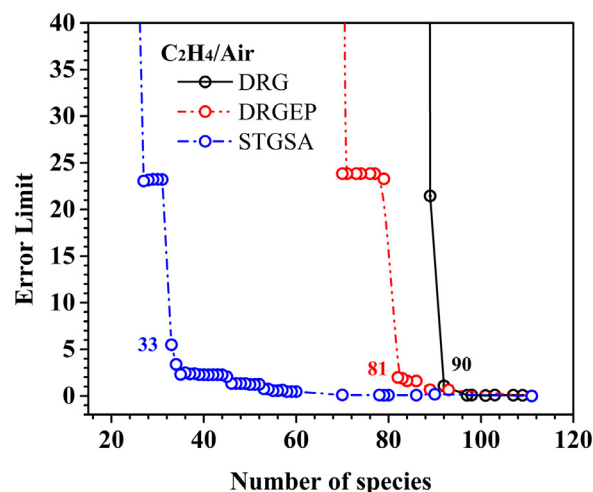
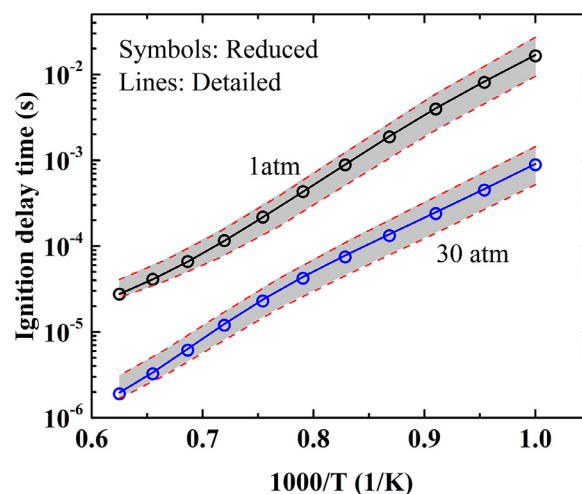
Figure 4 shows a list of the top 30 most important species under a wide range of operating conditions. The sensitivity index for a chemical species is measured by perturbing all reactions associated with such species. Consequently, a chemical species can be considered unimportant to the ignition delay time if the sensitivity index for such species is very small, and all reactions associated with such species are not important to the ignition delay time. Using STGSA, we can evaluate the impact of species on the ignition delay time, not just the impact of reactions on the ignition delay time. Moreover, since the importance of species can be evaluated, STGSA can be a powerful tool to mechanism reduction.

Table 2The total sensitivity indices of species for C_2H_4 /Air mixture under pressure of 1.0 atm, temperature of 1000 K and equivalence ratio of 1.0.

Species	Sample size					
	128	256	512	1024	2048	4098
HO_2	0.135	0.150	0.162	0.191	0.194	0.194
CH_2O	0.294	0.341	0.334	0.335	0.328	0.336
C_2H_3	0.825	0.856	0.824	0.823	0.812	0.82
$HCCO$	5.73×10^{-5}	5.40×10^{-5}	4.67×10^{-5}	4.81×10^{-5}	4.73×10^{-5}	4.69×10^{-5}
iC_3H_7	7.47×10^{-12}	7.46×10^{-12}	7.51×10^{-12}	7.75×10^{-12}	7.81×10^{-12}	7.78×10^{-12}

**Fig. 4.** The sensitivity indices for top 30 most important species under a wide range of operating conditions.

STGSA is employed to simplify the USC II mechanism of ethylene/air oxidation under a wide range of operating conditions and compared with the DRG-based methods. As seen in Fig. 5, STGSA can generate the smallest size of the reduced mechanism compared with DRG and DRGEP. When the maximum relative error is set at 20%, the reduced mechanism generated by STGSA only contains 33 species compared to the 82-species and the 90-species reduced mechanisms respectively generated by DRGEP and DRG. From Fig. 3, DRG-based methods cannot easily identify the global redundant species which are unimportant to ignition delay times under uncertainties in the reaction rates. However, STGSA can identify the global redundant species, leading to an optimal reduced mechanism under the input uncertainties. The computational load for STGSA is determined by the sample size, and it is usually higher than that for DRG-based methods. However, the computation time of STGSA can be greatly reduced using parallel computation because ignition delay times and sensitivity indices of species under different samples and different operating conditions can be computed independently. For combustion simulation, the simulation time depends linearly on the cubic of the number of species for a typical implicit method and calculation of chemical reactions almost may require 90% of the total computation time [52,53]. Thus, in mechanism reduction, achieving the more compact reduced mechanism is more important than the computation load spent by the reduction methods.

**Fig. 5.** Error limit as a function of number of species in the reduced mechanism for ethylene combustion under a wide range of initial temperatures (1000–1600 K), pressures (1–30 atm) and equivalence ratios (0.5–1.5).**Fig. 6.** The dependence of ignition delay time on the initial temperature under constant volume between the detailed and a 33-species reduced models for ethylene/air mixture at various pressures and equivalence ratio of 1.0. The grey shaded band indicates the 95.5% confidence intervals of probability density function of a 33-species reduced model uncertainty. The 95.5% confidence intervals of detailed model uncertainty by the red dash lines.

The ranking of the importance of species can be obtained using STGSA. Therefore, we can gradually reduce the size of the reaction mechanism and use the maximum relative error to evaluate the accuracy of the resulting reduced mechanisms. STGSA is found to generate the minimum size of the 33-species reduced mechanism with 20% error. Figure 6 shows the 33-species reduced mechanism can replicate the ignition delay times from the detailed mechanism at the nominal values of the rate parameters, and the uncertainty

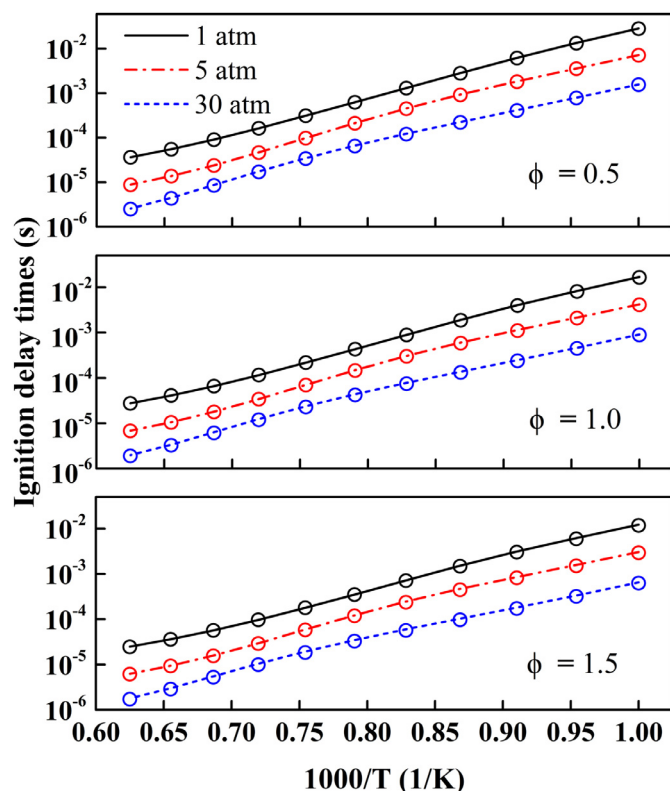


Fig. 7. The dependence of ignition delay time on the initial temperature under constant volume between the detailed, a 33-species reduced mechanism for ethylene/air mixture at various pressures and equivalence ratios.

bars of ignition delay times for the reduced mechanism are in good agreement with those for the detailed mechanism. Same results can be drawn under the different initial conditions. It is obvious that STGSA can accurately remove the global redundant species, and almost does not change the actual uncertainty bar, and original information of the detailed mechanism is highly retained for the reduced mechanism.

The reduced mechanisms are also checked for their accuracy in predicting the ignition delay times and laminar flame speeds under a wide range of operating conditions. Figure 7 shows ignition delay times as predicted by the detailed and 33-species reduced mechanisms generated by STGSA. It is seen that the 33-species reduced mechanism can accurately replicate the ignition delay times from the detailed mechanism, and the worst-case relative error in the ignition delay times between prediction of the reduced and detailed mechanisms is less than 6%. We also extend our comparison to some systems involving diffusive transport. Figure 8 shows the laminar flame speed predictions of the 33-species reduced mechanism generated by STGSA are in good agreement with those of the detailed mechanism. This result confirms that STGSA can identify the global redundant species and generate the reduced mechanism with very smaller size, and such reduced mechanism can retain the accurate predictability both for ignition delay times and laminar flame speeds.

3.3. Mechanisms reduction for *n*-heptane

n-Heptane is relevant in the studies of both gasoline and diesel fuels, and as such much attention has been paid to its mechanism construction and reduction. In this case, JetSurf (version 1.0) is used to calculate the auto-ignition of *n*-heptane/air mixtures for high-temperature application, and consisting of 194 species and 1459 reactions. STGSA is also applied to generate reduced mech-

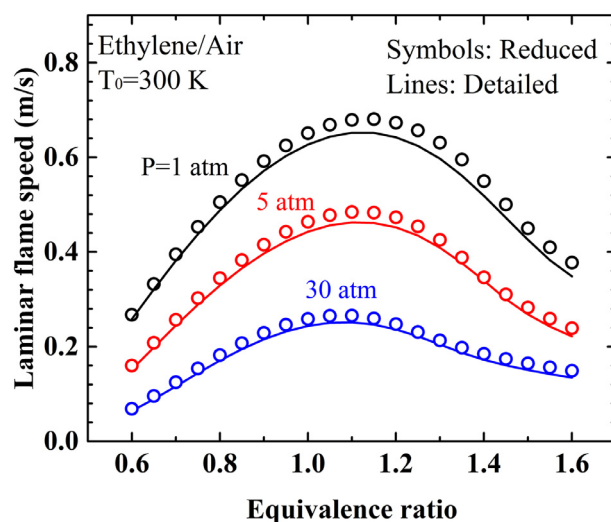


Fig. 8. Comparison of laminar flame speed between detailed mechanisms (lines) and the 33-species reduced mechanism (symbols) for pressure of 1 atm, 5 atm, and 30 atm under the unburned gas temperature of 300 K.

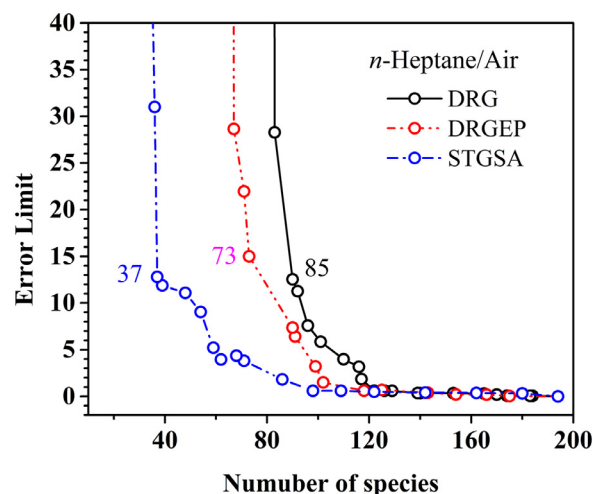


Fig. 9. Error limit as a function of number of species in the reduced mechanism for *n*-heptane combustion under a wide range of initial temperatures (1000–1600 K), pressures (1–30 atm) and equivalence ratios (0.5–1.5).

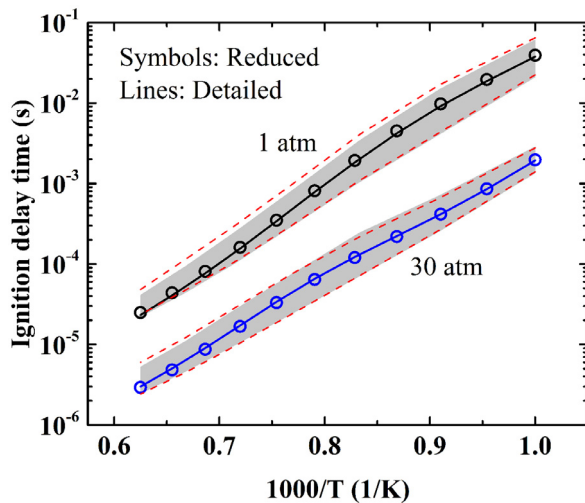
anism with smaller sizes. The minimum required sample size to achieve the correct ranking of the importance of species are shown in Table 3. We can obtain the reliable results of sensitivity indices for species with the sample size of 2048.

As shown in Fig. 9, the JetSurf mechanism is simplified by STGSA for the auto-ignition of *n*-heptane/air mixture and compared with DRG-based methods. It is seen that STGSA shows an accurate identification of the redundant species and can generate the smallest size of the reduced mechanism compared with DRG and DRGEP. When the maximum relative error is set at 20%, the reduced mechanism generated by STGSA only contains 37 species as compared to the 73-species and the 85-species reduced mechanisms respectively generated by DRGEP and DRG.

In this case, STGSA obtains the minimum size of the reduced mechanism with 37 species satisfying 20% error limit at the nominal values of the rate parameters, and the maximum error of predicting the ignition delay times is 15% at the pressure of 1 atm and equivalence ratios of 1.0. It is seen in Fig. 10 that the ignition delay times from the detailed mechanism can be replicated by the 37-species reduced mechanism with a minimal error. The uncertainty

Table 3The total sensitivity indices of some species for NC₇H₁₆/Air mixture using JetSurf mechanism under pressure of 1.0 atm, temperature of 1000 K and equivalence ratio of 1.0.

Species	Sample size					
	128	256	512	1024	2048	4098
HO ₂	0.725	0.861	0.871	0.887	0.906	0.907
CH ₃	0.637	0.658	0.699	0.727	0.740	0.739
HCO	0.0109	0.0127	0.0135	0.0137	0.0141	0.0140
C ₆ H ₅ O	4.72×10^{-14}	5.40×10^{-14}	4.76×10^{-14}	5.20×10^{-14}	5.28×10^{-14}	5.23×10^{-14}
C ₆ H ₄ CH ₃	4.39×10^{-14}	5.39×10^{-14}	4.70×10^{-14}	4.67×10^{-14}	4.69×10^{-14}	4.72×10^{-14}

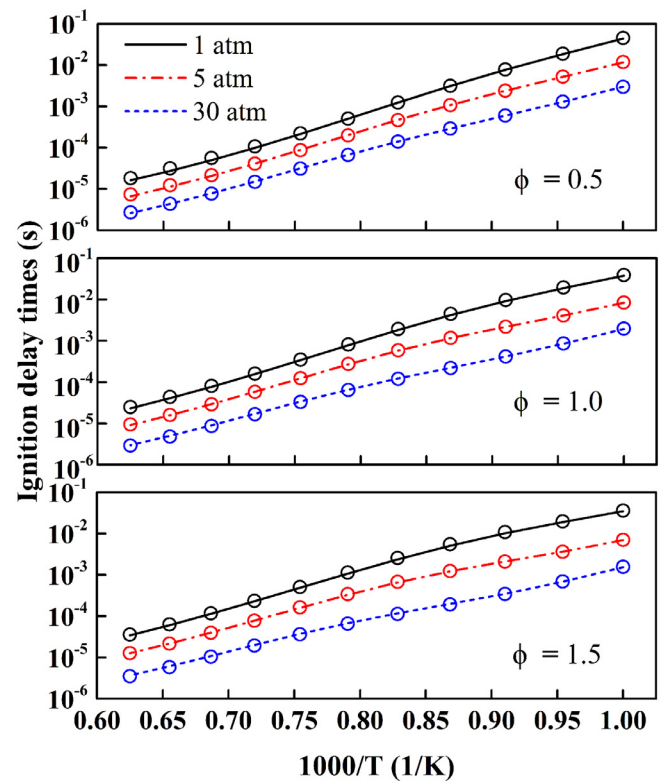
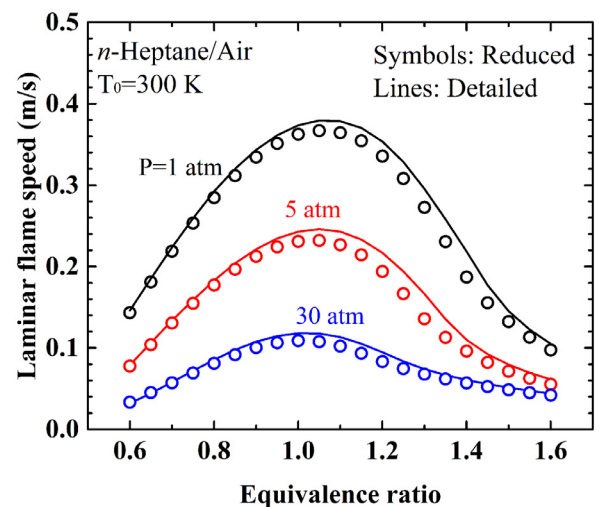
**Fig. 10.** The dependence of ignition delay time on the initial temperature under constant volume between the detailed and a 37-species reduced models for n-heptane/air mixture at various pressures and equivalence ratio of 1.0. The grey shaded band indicates the 95.5% confidence intervals of probability density function of a 33-species reduced model uncertainty. The 95.5% confidence intervals of detailed model uncertainty by the red dash lines.

bars of ignition delay times induced by the uncertainties in the reaction rates for the reduced mechanism are agreement with that for the detailed mechanism. However, the visible errors can be observed in Fig. 10 under the pressure of 1.0 atm because of the 15% error of predicting the ignition delay times for the reduced mechanism. If there is no any visible error in Fig. 10, the error limit should be set less than 10%.

In Fig. 11, the ignition delay times predicted using the 37-species reduced mechanism generated by STGSA are compared to those predicted using the detailed mechanism. It is seen that the 37-species can accurately replicate the ignition delay times of the detailed mechanism, and the worst-case relative error is less than 16%. The 37-species reduced mechanism is also used to predict the laminar flame speeds which are not part of the reduction process. Figure 12 shows the laminar flame speeds prediction of the 37-species reduced mechanism agree well with those of the detailed mechanism, with the maximum relative error being less than 19%. Two cases of reducing chemical mechanisms show that STGSA can identify the global redundant species and generate the reliable reduced mechanism under uncertainties in the reaction rates. Two reduced mechanisms for ethylene/air and n-heptane/air are given in the Supplemental Material.

4. Conclusions

Uncertainties in the reaction rate coefficients can greatly impact the contribution of the production rate of one species on another. This leads to the essential problem of identifying important species using DRG-based methods. Through uncertainty analysis of

**Fig. 11.** The dependence of ignition delay time on the initial temperature under constant volume between the detailed, a 37-species reduced mechanism for n-heptane/air mixture at various pressures and equivalence ratios.**Fig. 12.** Comparison of laminar flame speed between detailed mechanisms (lines) and the 37-species reduced mechanism (symbols) for pressure of 1 atm, 5 atm, and 30 atm under the unburned gas temperature of 300 K.

the coupling species identified by DRGEP using USC II mechanism for ethylene/air oxidation, it is shown that the uncertainties in the reaction rate coefficients can greatly impact the size of the resulting reduced mechanism and identification of the important species. Consequently, in order to achieve a reliable skeletal mechanism, STGSA is proposed to evaluate the importance of the chemical species with consideration of uncertainties in the reaction rate coefficients. This global reduction method is then applied to simplify ethylene oxidation of the USC II mechanism consisting of 111 species and n-heptane oxidation of the JetSurf (version 1.0) mechanism consisting of 194 species. The resulting reduced mechanisms for the ethylene and the n-heptane oxidation involve only 33 species and 37 species, respectively, and such sizes of the resulting reduced mechanisms are substantially smaller than those of the reduced mechanisms simplified by DRG-based methods. Furthermore, it is demonstrated that the resulting reduced mechanisms can mimic the performance of the detailed mechanism for the ignition delay times as well as the laminar flame speeds.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2020.12.004](https://doi.org/10.1016/j.combustflame.2020.12.004).

References

- [1] C.K. Westbrook, Y. Mizobuchi, T.J. Poinot, P.J. Smith, J. Warnatz, Computational combustion, *Proc. Combust. Inst.* 30 (2005) 125–157.
- [2] S.L. Fischer, F.L. Dryer, H.J. Curran, The reaction kinetics of dimethyl ether. I: high-temperature pyrolysis and oxidation in flow reactors, *Int. J. Chem. Kinet.* 32 (2000) 713–740.
- [3] K. Kumar, C.J. Sung, X. Hui, Laminar flame speeds and extinction limits of conventional and alternative jet fuels, *Fuel* 90 (2011) 1004–1011.
- [4] T. Turányi, A.S. Tomlin, *Analysis of Kinetic Reaction Mechanisms*, Springer Berlin Heidelberg, 2014.
- [5] T.F. Lu, C.K. Law, A directed relation graph method for mechanism reduction, *Proc. Combust. Inst.* 30 (2005) 1333–1341.
- [6] F. Battin-Leclerc, J.M. Simmie, E. Blurock, *Cleaner Combustion*, Springer, 2013.
- [7] T. Turányi, Sensitivity analysis of complex kinetic systems. Tools and applications, *J. Math. Chem.* 5 (1990) 203–248.
- [8] Y. Ju, T. Niioka, Reduced kinetic mechanism of ignition for nonpremixed hydrogen/air in a supersonic mixing layer, *Combust. Flame* 99 (1994) 240–246.
- [9] G. Esposito, H.K. Chelliah, Skeletal reaction models based on principal component analysis: application to ethylene–air ignition, propagation, and extinction phenomena, *Combust. Flame* 158 (2012) 477–489.
- [10] N.J. Brown, G. Li, M.L. Koszykowski, Mechanism reduction via principal component analysis, *Int. J. Chem. Kinet.* 29 (1997) 393–414.
- [11] M. Xu, Y. Fan, J. Yuan, Simplification of the mechanism of NO_x formation in a CH₄/air combustion system, *Int. J. Energy. Res.* 23 (1999) 1267–1276.
- [12] T. Lu, C.K. Law, On the applicability of directed relation graphs to the reduction of reaction mechanisms, *Combust. Flame* 146 (2006) 472–483.
- [13] T. Lu, C.K. Law, A criterion based on computational singular perturbation for the identification of quasi steady state species: a reduced mechanism for methane oxidation with NO chemistry, *Combust. Flame* 154 (2008) 761–774.
- [14] W. Sun, Z. Chen, X. Gou, Y. Ju, A path flux analysis method for the reduction of detailed chemical kinetic mechanisms, *Combust. Flame* 157 (2010) 1298–1307.
- [15] X. Gou, W. Sun, Z. Chen, Y. Ju, A dynamic multi-timescale method for combustion modeling with detailed and reduced chemical kinetic mechanisms, *Combust. Flame* 157 (2010) 1111–1121.
- [16] X. Gao, S. Yang, W. Sun, A global pathway selection algorithm for the reduction of detailed chemical kinetic mechanisms, *Combust. Flame* 167 (2016) 238–247.
- [17] S. Lam, D. Goussis, The CSP method for simplifying kinetics, *Int. J. Chem. Kinet.* 26 (1994) 461–486.
- [18] P. Koniaritis, S. Rigopoulos, W. Jones, A methodology for derivation of RCCE-reduced mechanisms via CSP, *Combust. Flame* 183 (2017) 126–143.
- [19] P. Koniaritis, S. Rigopoulos, W. Jones, Reduction of a detailed chemical mechanism for a kerosene surrogate via RCCE-CSP, *Combust. Flame* 194 (2018) 85–106.
- [20] R.M. Galassi, M. Valorani, H.N. Najm, C. Safta, M. Khalil, P.P. Ciottoli, Chemical model reduction under uncertainty, *Combust. Flame* 179 (2017) 242–252.
- [21] R.M. Galassi, P.P. Ciottoli, S.M. Sarathy, H.G. Im, S. Paolucci, M. Valorani, Automated chemical kinetic mechanism simplification with minimal user expertise, *Combust. Flame* 197 (2018) 439–448.
- [22] X.L. Zheng, T.F. Lu, C.K. Law, Experimental counterflow ignition temperatures and reaction mechanisms of 1,3-butadiene, *Proc. Combust. Inst.* 31 (2007) 367–375.
- [23] Z.Y. Luo, T.F. Lu, J.W. Liu, A reduced mechanism for ethylene/methane mixtures with excessive NO enrichment, *Combust. Flame* 158 (2011) 1245–1254.
- [24] K.E. Niemeyer, C.J. Sung, M.P. Raju, Skeletal mechanism generation for surrogate fuels using directed relation graph with error propagation and sensitivity analysis, *Combust. Flame* 157 (2016) 1760–1770.
- [25] T. Turányi, Kinal – a program package for kinetic-analysis of reaction-mechanisms, *Comput. Chem.* 14 (1990) 253–254.
- [26] R. Li, S.H. Li, F. Wang, X.Y. Li, Sensitivity analysis based on intersection approach for mechanism reduction of cyclohexane, *Combust. Flame* 166 (2016) 55–65.
- [27] Y. Chang, M. Jia, B. Niu, Y. Zhang, M. Xie, Y. Li, Construction and assessment of reduced oxidation mechanisms using global sensitivity analysis and uncertainty analysis, *Proc. Combust. Inst.* 37 (2019) 751–761.
- [28] Y. Chang, M. Jia, B. Niu, X. Dong, P. Wang, Reduction of large-scale chemical mechanisms using global sensitivity analysis on reaction class/sub-mechanism, *Combust. Flame* 212 (2020) 355–366.
- [29] M.J. Davis, R.T. Skodje, A.S. Tomlin, Global sensitivity analysis of chemical-kinetic reaction mechanisms: construction and deconstruction of the probability density function, *J. Phys. Chem. A* 115 (2011) 1556–1578.
- [30] M.J. Davis, W. Liu, R. Sivaramakrishnan, Global sensitivity analysis with small sample sizes: ordinary least squares approach, *J. Phys. Chem. A* 121 (2017) 553–570.
- [31] D.D. Zhou, M.J. Davis, R.T. Skodje, Multitarget global sensitivity analysis of n-butanol combustion, *J. Phys. Chem. A* 117 (2013) 3569–3584.
- [32] M. Hantouche, S.M. Sarathy, O.M. Knio, Global sensitivity analysis of n-butanol reaction kinetics using rate rules, *Combust. Flame* 196 (2018) 452–465.
- [33] T. Ziehn, A.S. Tomlin, GUI-HDMR – a software tool for global sensitivity analysis of complex models, *Environ. Modell. Softw.* 24 (2009) 775–785.
- [34] J. An, G. He, F. Qin, R. Li, Z. Huang, A new framework of global sensitivity analysis for the chemical kinetic model using PSO-BPNN, *Comput. Chem. Eng.* 112 (2018) 154–164.
- [35] A. Saltelli, R. Bolado, An alternative way to compute Fourier amplitude sensitivity test (FAST), *Comput. Stat. Data An.* 26 (1998) 445–460.
- [36] S. Li, B. Yang, F. Qi, Accelerate global sensitivity analysis using artificial neural network algorithm: case studies for combustion kinetic model, *Combust. Flame* 168 (2016) 53–64.
- [37] V. Todorov, I. Dimov, T. Ostromsky, S. Apostolov, R. Georgieva, Y. Dimitrov, Z. Zlatev, Advanced stochastic approaches for Sobol'sensitivity indices evaluation, *Neural Comput. Appl.* (2020) 1–16.
- [38] L.F. Nurislamova, I.M. Gubaydullin, Research and reduction of mathematical model of chemical reaction by Sobol method, *Comput. Res. Model.* 8 (2016) 633–646.
- [39] L.F. Nurislamova, I.M. Gubaydullin, Mechanism reduction of chemical reaction based on sensitivity analysis: development and testing of some new procedure, *J. Math. Chem.* 55 (2017) 1–14.
- [40] M.S. Okino, M.L. Mavrovouniotis, Simplification of mathematical models of chemical reaction systems, *Chem. Rev.* 98 (1999) 391–408.
- [41] I.M. Sobol, Sensitivity estimates for nonlinear mathematical models, *Matem. Mod.* 1 (1993) 407–414.
- [42] A. Saltelli, M. Ratto, S. Tarantola, F. Campolongo, Sensitivity analysis for chemical models, *Chem. Rev.* 105 (2005) 2811–2828.
- [43] H. Wang, D.A. Sheen, Combustion kinetic model uncertainty quantification, propagation and minimization, *Prog. Energy Combust.* 47 (2014) 1–31.
- [44] J. Wang, S. Li, B. Yang, Combustion kinetic model development using surrogate model similarity method, *Combust. Theor. Model.* 22 (2018) 777–794.
- [45] A. Saltelli, P. Annoni, I. Azzini, F. Campolongo, M. Ratto, S. Tarantola, Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index, *Comput. Phys. Commun.* 181 (2010) 259–270.
- [46] D.L. Baulch, C.J. Cobos, R.A. Cox, C. Esser, P. Frank, T. Just, J.A. Kerr, M.J. Pilling, J. Troe, R.W. Walker, J. Warnatz, Evaluated kinetic data for combustion modeling, *J. Phys. Chem. Ref. Data* 21 (1992) 411–734.
- [47] D.L. Baulch, C.T. Bowman, C.J. Cobos, R.A. Cox, T. Just, J.A. Kerr, M.J. Pilling, D. Stocker, J. Troe, W. Tsang, R.W. Walker, J. Warnatz, Evaluated kinetic data for combustion modeling: supplement II, *J. Phys. Chem. Ref. Data* 34 (2005) 757–1397.

- [48] T. Lu, C.K. Law, Toward accommodating realistic fuel chemistry in large-scale computations, *Prog. Energy Combust.* 35 (2009) 192–215.
- [49] H. Wang, X. You, A.V. Joshi, S.G. Davis, A. Laskin, F. Egolfopoulos and C.K. Law, USC mech version II. High-temperature combustion reaction model of H₂/CO/C₁-C₄ compounds, http://ignis.usc.edu/USC_Mech_II.htm, 2007.
- [50] B. Sirjean, E. Dames, D.A. Sheen, X. You, C. Sung, A.T. Holley, F.N. Egolfopoulos, H. Wang, S.S. Vasu, D.F. Davidson, R.K. Hanson, H. Pitsch, C.T. Bowman, A. Kelley, C.K. Law, W. Tsang, N.P. Cernansky, D.L. Miller, A. Violi and R.P. Lindstedt, A high-temperature chemical kinetic model of n-alkane oxidation, Jet-SurF version 1.0, <http://web.stanford.edu/group/haiwanglab/JetSurF/JetSurF1.0/index.html>, 2009.
- [51] D.A. Sheen, X.Q. You, H. Wang, T. Lovas, Spectral uncertainty quantification, propagation and optimization of a detailed kinetic model for ethylene combustion, *Proc. Combust. Inst.* 32 (2009) 535–542.
- [52] M. Lamnaouer, R.C. Ryder, A. Brankovic, E.L. Petersen, Reduced combustion time model for methane in gas turbine flow fields, *J. Nat. Gas. Chem.* 18 (2009) 145–155.
- [53] B. Bhattacharjee, D.A. Schwer, P.I. Barton, W.H. Green, Optimally-reduced kinetic models: reaction elimination in large-scale kinetic mechanisms, *Combust. Flame* 135 (2003) 191–208.